

# Mathematical Foundations of Machine Learning

## Assignment 6 Solutions

### Question 1 (15 points)

Let  $\mathbf{A} \in \mathbb{R}^{n \times n}$  be symmetric positive definite. Suppose  $\mathbf{A}$  admits a Cholesky factorization

$$\mathbf{A} = \mathbf{L}\mathbf{L}^\top,$$

where  $\mathbf{L} = (\ell_{ij})$  is lower-triangular with strictly positive diagonal entries  $\ell_{11}, \dots, \ell_{nn}$ .

(a) (5 points) Prove the following entry-wise formulas for  $\mathbf{L}$ :

$$\ell_{ii} = \sqrt{a_{ii} - \sum_{k=1}^{i-1} \ell_{ik}^2} \quad (i = 1, \dots, n),$$

and for  $i > j$ ,

$$\ell_{ij} = \frac{1}{\ell_{jj}} \left( a_{ij} - \sum_{k=1}^{j-1} \ell_{ik} \ell_{jk} \right).$$

(b) (6 points) Prove uniqueness: if

$$\mathbf{A} = \mathbf{L}\mathbf{L}^\top = \tilde{\mathbf{L}}\tilde{\mathbf{L}}^\top$$

with  $\mathbf{L}, \tilde{\mathbf{L}}$  both lower-triangular and both having strictly positive diagonal entries, then  $\mathbf{L} = \tilde{\mathbf{L}}$ .

(c) (4 points) Compute the Cholesky factor  $\mathbf{L}$  and  $\det(\mathbf{A})$  for

$$\mathbf{A} = \begin{bmatrix} 4 & 2 & -2 \\ 2 & 10 & 5 \\ -2 & 5 & 9 \end{bmatrix}.$$

**Solution.**

(a) Since  $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$  and  $\mathbf{L}$  is lower-triangular,

$$a_{ij} = (\mathbf{L}\mathbf{L}^\top)_{ij} = \sum_{k=1}^n \ell_{ik} \ell_{jk} = \sum_{k=1}^{\min(i,j)} \ell_{ik} \ell_{jk}.$$

For  $i = j$ ,

$$a_{ii} = \sum_{k=1}^i \ell_{ik}^2 = \left( \sum_{k=1}^{i-1} \ell_{ik}^2 \right) + \ell_{ii}^2,$$

hence

$$\ell_{ii}^2 = a_{ii} - \sum_{k=1}^{i-1} \ell_{ik}^2.$$

Because  $\ell_{ii} > 0$ , taking the positive square root yields

$$\ell_{ii} = \sqrt{a_{ii} - \sum_{k=1}^{i-1} \ell_{ik}^2}.$$

For  $i > j$ ,

$$a_{ij} = \sum_{k=1}^j \ell_{ik} \ell_{jk} = \left( \sum_{k=1}^{j-1} \ell_{ik} \ell_{jk} \right) + \ell_{ij} \ell_{jj}.$$

Rearranging and dividing by  $\ell_{jj} > 0$  gives

$$\ell_{ij} = \frac{1}{\ell_{jj}} \left( a_{ij} - \sum_{k=1}^{j-1} \ell_{ik} \ell_{jk} \right).$$

- (b) Assume  $\mathbf{A} = \mathbf{L}\mathbf{L}^\top = \tilde{\mathbf{L}}\tilde{\mathbf{L}}^\top$  with  $\mathbf{L}, \tilde{\mathbf{L}}$  lower-triangular and with positive diagonal. Compare the  $(1, 1)$  entry:

$$a_{11} = \ell_{11}^2 = \tilde{\ell}_{11}^2 \Rightarrow \ell_{11} = \tilde{\ell}_{11} \quad (\text{since both are } > 0).$$

For  $i \geq 2$ , compare  $(i, 1)$ :

$$a_{i1} = \ell_{i1} \ell_{11} = \tilde{\ell}_{i1} \tilde{\ell}_{11} \Rightarrow \ell_{i1} = \tilde{\ell}_{i1}.$$

Now assume inductively that columns  $1, 2, \dots, j-1$  of  $\mathbf{L}$  and  $\tilde{\mathbf{L}}$  agree for some  $2 \leq j \leq n$ . Compare  $(j, j)$ :

$$a_{jj} = \sum_{k=1}^j \ell_{jk}^2 = \left( \sum_{k=1}^{j-1} \ell_{jk}^2 \right) + \ell_{jj}^2, \quad a_{jj} = \left( \sum_{k=1}^{j-1} \tilde{\ell}_{jk}^2 \right) + \tilde{\ell}_{jj}^2.$$

By the induction hypothesis, the partial sums match, so  $\ell_{jj}^2 = \tilde{\ell}_{jj}^2$ , and positivity implies  $\ell_{jj} = \tilde{\ell}_{jj}$ . For any  $i > j$ , compare  $(i, j)$ :

$$a_{ij} = \sum_{k=1}^j \ell_{ik} \ell_{jk} = \left( \sum_{k=1}^{j-1} \ell_{ik} \ell_{jk} \right) + \ell_{ij} \ell_{jj},$$

and similarly

$$a_{ij} = \left( \sum_{k=1}^{j-1} \tilde{\ell}_{ik} \tilde{\ell}_{jk} \right) + \tilde{\ell}_{ij} \tilde{\ell}_{jj}.$$

Using the induction hypothesis and  $\ell_{jj} = \tilde{\ell}_{jj} > 0$  gives  $\ell_{ij} = \tilde{\ell}_{ij}$ . Hence column  $j$  matches. By induction,  $\mathbf{L} = \tilde{\mathbf{L}}$ .

- (c) Apply the formulas from (a).

$$\ell_{11} = \sqrt{4} = 2, \quad \ell_{21} = \frac{2}{2} = 1, \quad \ell_{31} = \frac{-2}{2} = -1.$$

Next,

$$\ell_{22} = \sqrt{10 - \ell_{21}^2} = \sqrt{10 - 1} = 3, \quad \ell_{32} = \frac{5 - \ell_{31}\ell_{21}}{\ell_{22}} = \frac{5 - (-1)(1)}{3} = 2.$$

Finally,

$$\ell_{33} = \sqrt{9 - \ell_{31}^2 - \ell_{32}^2} = \sqrt{9 - 1 - 4} = 2.$$

Therefore,

$$\mathbf{L} = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ -1 & 2 & 2 \end{bmatrix}.$$

Since  $\det(\mathbf{A}) = \det(\mathbf{L})^2$  and  $\det(\mathbf{L}) = \ell_{11}\ell_{22}\ell_{33}$ ,

$$\det(\mathbf{A}) = (2 \cdot 3 \cdot 2)^2 = 144.$$

## Question 2 (20 points)

Let

$$\mathbf{B} = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}, \quad \mathbf{v}_0 = \begin{bmatrix} -1 \\ -4 \\ 3 \end{bmatrix}, \quad \mathbf{v}_{t+1} = \mathbf{B}\mathbf{v}_t \quad (t = 0, 1, 2, \dots).$$

- (a) (6 points) Compute all eigenvalues of  $\mathbf{B}$  and, for each eigenvalue, find a corresponding eigenvector.
- (b) (4 points) Show that  $\mathbf{B}$  is diagonalizable and find an invertible matrix  $\mathbf{S}$  and a diagonal matrix  $\mathbf{\Lambda}$  such that

$$\mathbf{\Lambda} = \mathbf{S}^{-1}\mathbf{B}\mathbf{S}.$$

- (c) (5 points) Use your diagonalization to obtain a closed form for  $\mathbf{v}_t$  and compute  $\mathbf{v}_5$ .
- (d) (5 points) General fact (use the notation from the slides). Suppose  $\mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  for some invertible  $\mathbf{P}$  and diagonal  $\mathbf{D}$ , and assume  $\mathbf{D}$  is invertible. Prove that

$$\mathbf{A}^{-1} = \mathbf{P}\mathbf{D}^{-1}\mathbf{P}^{-1}.$$

Then compute  $\mathbf{B}^{-1}\mathbf{v}_0$  for the given  $\mathbf{B}$  (do not use Gaussian elimination).

**Solution.**

- (a) The characteristic polynomial is

$$p_{\mathbf{B}}(\lambda) = \det(\mathbf{B} - \lambda\mathbf{I}) = \det \begin{bmatrix} -\lambda & 0 & -2 \\ 1 & -\lambda & 1 \\ 0 & 1 & 2 - \lambda \end{bmatrix}.$$

Expand along the first row:

$$\begin{aligned} p_{\mathbf{B}}(\lambda) &= (-\lambda) \det \begin{bmatrix} -\lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix} + (-2) \det \begin{bmatrix} 1 & -\lambda \\ 0 & 1 \end{bmatrix} \\ &= (-\lambda)((-\lambda)(2 - \lambda) - 1) - 2 \\ &= -\lambda^3 + 2\lambda^2 + \lambda - 2. \end{aligned}$$

Roots are unchanged by multiplying by  $-1$ , so solve

$$\lambda^3 - 2\lambda^2 - \lambda + 2 = 0.$$

Checking  $\lambda = 1$  gives 0, so  $(\lambda - 1)$  is a factor. Division yields

$$\lambda^3 - 2\lambda^2 - \lambda + 2 = (\lambda - 1)(\lambda^2 - \lambda - 2) = (\lambda - 1)(\lambda - 2)(\lambda + 1).$$

Hence eigenvalues are  $\lambda = 1, 2, -1$ .

Eigenvectors:

- For  $\lambda = 1$ , solve  $(\mathbf{B} - \mathbf{I})\mathbf{x} = \mathbf{0}$ :

$$\mathbf{B} - \mathbf{I} = \begin{bmatrix} -1 & 0 & -2 \\ 1 & -1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \Rightarrow \mathbf{x} = x_3 \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix}.$$

$$\text{Take } \mathbf{q}_1 = \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix}.$$

- For  $\lambda = -1$ , solve  $(\mathbf{B} + \mathbf{I})\mathbf{x} = \mathbf{0}$ :

$$\mathbf{B} + \mathbf{I} = \begin{bmatrix} 1 & 0 & -2 \\ 1 & 1 & 1 \\ 0 & 1 & 3 \end{bmatrix} \Rightarrow \mathbf{x} = x_3 \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}.$$

Take  $\mathbf{q}_2 = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}$ .

- For  $\lambda = 2$ , solve  $(\mathbf{B} - 2\mathbf{I})\mathbf{x} = \mathbf{0}$ :

$$\mathbf{B} - 2\mathbf{I} = \begin{bmatrix} -2 & 0 & -2 \\ 1 & -2 & 1 \\ 0 & 1 & 0 \end{bmatrix} \Rightarrow \mathbf{x} = x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Take  $\mathbf{q}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ .

(b) Since  $\mathbf{B}$  has three distinct eigenvalues, it is diagonalizable. Let

$$\mathbf{S} = [\mathbf{q}_1 \quad \mathbf{q}_2 \quad \mathbf{q}_3] = \begin{bmatrix} -2 & 2 & -1 \\ -1 & -3 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad \mathbf{\Lambda} = \text{diag}(1, -1, 2).$$

Then  $\mathbf{BS} = \mathbf{S}\mathbf{\Lambda}$ , hence  $\mathbf{S}^{-1}\mathbf{BS} = \mathbf{\Lambda}$ .

(c) Decompose  $\mathbf{v}_0$  in the eigenbasis:

$$\mathbf{v}_0 = \begin{bmatrix} -1 \\ -4 \\ 3 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \mathbf{q}_1 + \mathbf{q}_2 + \mathbf{q}_3.$$

Since  $\mathbf{v}_t = \mathbf{B}^t \mathbf{v}_0$  and  $\mathbf{B}^t \mathbf{q}_i = \lambda_i^t \mathbf{q}_i$ ,

$$\mathbf{v}_t = 1^t \mathbf{q}_1 + (-1)^t \mathbf{q}_2 + 2^t \mathbf{q}_3 = \mathbf{q}_1 + (-1)^t \mathbf{q}_2 + 2^t \mathbf{q}_3.$$

For  $t = 5$ ,

$$\mathbf{v}_5 = \mathbf{q}_1 - \mathbf{q}_2 + 32 \mathbf{q}_3 = \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} + 32 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -36 \\ 2 \\ 32 \end{bmatrix}.$$

(d) Since  $\mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ , multiply by  $\mathbf{P}$  on the left and  $\mathbf{P}^{-1}$  on the right to get

$$\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}.$$

If  $\mathbf{D}$  is invertible, then

$$\mathbf{A}^{-1} = (\mathbf{P}\mathbf{D}\mathbf{P}^{-1})^{-1} = \mathbf{P}\mathbf{D}^{-1}\mathbf{P}^{-1},$$

using  $(\mathbf{XYZ})^{-1} = \mathbf{Z}^{-1}\mathbf{Y}^{-1}\mathbf{X}^{-1}$ .

For  $\mathbf{B}$ , we use the expansion  $\mathbf{v}_0 = \mathbf{q}_1 + \mathbf{q}_2 + \mathbf{q}_3$  and invert eigenvalues:

$$\mathbf{B}^{-1}\mathbf{v}_0 = 1^{-1}\mathbf{q}_1 + (-1)^{-1}\mathbf{q}_2 + 2^{-1}\mathbf{q}_3 = \mathbf{q}_1 - \mathbf{q}_2 + \frac{1}{2}\mathbf{q}_3.$$

Thus

$$\mathbf{B}^{-1}\mathbf{v}_0 = \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{9}{2} \\ 2 \\ \frac{1}{2} \end{bmatrix}.$$

### Question 3 (20 points)

Let

$$\mathbf{A} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \\ 0 & 2 \\ 0 & 0 \end{bmatrix} \in \mathbb{R}^{4 \times 2}.$$

- (a) (6 points) Compute  $\mathbf{A}^\top \mathbf{A}$  and find its eigenvalues and an orthonormal eigenbasis  $\{\mathbf{v}_1, \mathbf{v}_2\}$ .
- (b) (4 points) Use part (a) to determine the singular values  $\sigma_1 \geq \sigma_2 > 0$  of  $\mathbf{A}$ .
- (c) (10 points) Construct an SVD  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$  with  $\mathbf{U} \in \mathbb{R}^{4 \times 4}$  orthogonal,  $\mathbf{V} \in \mathbb{R}^{2 \times 2}$  orthogonal, and  $\mathbf{\Sigma} \in \mathbb{R}^{4 \times 2}$  diagonal (with nonnegative diagonal entries). Your final answer must give explicit  $\mathbf{U}, \mathbf{\Sigma}, \mathbf{V}$ .

**Solution.**

(a)

$$\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \\ 0 & 2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 1 & 5 \end{bmatrix}.$$

The eigenvalues of  $\begin{bmatrix} 5 & 1 \\ 1 & 5 \end{bmatrix}$  are 6 and 4. An orthonormal eigenbasis is

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

(b) Singular values satisfy  $\sigma_i^2 = \lambda_i(\mathbf{A}^\top \mathbf{A})$ , hence

$$\sigma_1 = \sqrt{6}, \quad \sigma_2 = 2.$$

(c) Let

$$\mathbf{V} = [\mathbf{v}_1 \quad \mathbf{v}_2] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Define  $\mathbf{u}_i = \frac{1}{\sigma_i} \mathbf{A} \mathbf{v}_i$  for  $i = 1, 2$ . First,

$$\mathbf{A} \mathbf{v}_1 = \mathbf{A} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 2 \\ 2 \\ 2 \\ 0 \end{bmatrix}, \quad \mathbf{u}_1 = \frac{1}{\sqrt{6}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 2 \\ 2 \\ 2 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}.$$

Next,

$$\mathbf{A} \mathbf{v}_2 = \mathbf{A} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 2 \\ 0 \\ -2 \\ 0 \end{bmatrix}, \quad \mathbf{u}_2 = \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 2 \\ 0 \\ -2 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}.$$

Complete  $\{\mathbf{u}_1, \mathbf{u}_2\}$  to an orthonormal basis of  $\mathbb{R}^4$  by choosing

$$\mathbf{u}_3 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{u}_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

Set

$$\mathbf{U} = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \mathbf{u}_3 \quad \mathbf{u}_4], \quad \mathbf{\Sigma} = \begin{bmatrix} \sqrt{6} & 0 \\ 0 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Then  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$  is an SVD.

## Question 4 (15 points)

Let  $\mathbf{A} \in \mathbb{R}^{m \times n}$  and let  $\lambda > 0$ . Define

$$E_\lambda := \{\mathbf{v} \in \mathbb{R}^n : \mathbf{A}^\top \mathbf{A} \mathbf{v} = \lambda \mathbf{v}\}, \quad F_\lambda := \{\mathbf{u} \in \mathbb{R}^m : \mathbf{A} \mathbf{A}^\top \mathbf{u} = \lambda \mathbf{u}\}.$$

- (a) (5 points) Show that if  $\mathbf{v} \in E_\lambda$ , then  $\mathbf{A} \mathbf{v} \in F_\lambda$  and

$$\frac{1}{\lambda} \mathbf{A}^\top (\mathbf{A} \mathbf{v}) = \mathbf{v}.$$

- (b) (5 points) Show that if  $\mathbf{u} \in F_\lambda$ , then  $\mathbf{A}^\top \mathbf{u} \in E_\lambda$  and

$$\frac{1}{\lambda} \mathbf{A} (\mathbf{A}^\top \mathbf{u}) = \mathbf{u}.$$

- (c) (5 points) Deduce that the *multiset* of nonzero eigenvalues of  $\mathbf{A}^\top \mathbf{A}$  equals the multiset of nonzero eigenvalues of  $\mathbf{A} \mathbf{A}^\top$  (including multiplicities). Here, “multiset” means that eigenvalues are counted with repetition according to multiplicity.

In addition, show the following correspondence: if  $\mathbf{v}$  is an eigenvector of  $\mathbf{A}^\top \mathbf{A}$  with eigenvalue  $\lambda > 0$ , then  $\mathbf{A} \mathbf{v}$  is an eigenvector of  $\mathbf{A} \mathbf{A}^\top$  with eigenvalue  $\lambda$ .

**Solution.**

- (a) Let  $\mathbf{v} \in E_\lambda$ . Then

$$\mathbf{A} \mathbf{A}^\top (\mathbf{A} \mathbf{v}) = \mathbf{A} (\mathbf{A}^\top \mathbf{A} \mathbf{v}) = \mathbf{A} (\lambda \mathbf{v}) = \lambda (\mathbf{A} \mathbf{v}),$$

so  $\mathbf{A} \mathbf{v} \in F_\lambda$ .

Also,

$$\frac{1}{\lambda} \mathbf{A}^\top (\mathbf{A} \mathbf{v}) = \frac{1}{\lambda} \mathbf{A}^\top \mathbf{A} \mathbf{v} = \frac{1}{\lambda} (\lambda \mathbf{v}) = \mathbf{v}.$$

- (b) Let  $\mathbf{u} \in F_\lambda$ . Then

$$\mathbf{A}^\top \mathbf{A} (\mathbf{A}^\top \mathbf{u}) = \mathbf{A}^\top (\mathbf{A} \mathbf{A}^\top \mathbf{u}) = \mathbf{A}^\top (\lambda \mathbf{u}) = \lambda (\mathbf{A}^\top \mathbf{u}),$$

so  $\mathbf{A}^\top \mathbf{u} \in E_\lambda$ .

Also,

$$\frac{1}{\lambda} \mathbf{A} (\mathbf{A}^\top \mathbf{u}) = \frac{1}{\lambda} \mathbf{A} \mathbf{A}^\top \mathbf{u} = \frac{1}{\lambda} (\lambda \mathbf{u}) = \mathbf{u}.$$

- (c) Let  $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$  be a basis of  $E_\lambda$ . We claim that

$$\{\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_k\}$$

is a basis of  $F_\lambda$ .

First, these vectors are linearly independent. Indeed, if

$$c_1 \mathbf{A} \mathbf{v}_1 + \dots + c_k \mathbf{A} \mathbf{v}_k = \mathbf{0},$$

then

$$\mathbf{A} (c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k) = \mathbf{0}.$$

Let

$$\mathbf{v} = c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k \in E_\lambda.$$

By part (a),

$$\mathbf{v} = \frac{1}{\lambda} \mathbf{A}^\top (\mathbf{A} \mathbf{v}) = \frac{1}{\lambda} \mathbf{A}^\top \mathbf{0} = \mathbf{0}.$$

Since  $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$  is linearly independent, all  $c_i = 0$ .

Next, they span  $F_\lambda$ . Let  $\mathbf{u} \in F_\lambda$ . By part (b),

$$\mathbf{A}^\top \mathbf{u} \in E_\lambda.$$

Hence there exist scalars  $c_1, \dots, c_k$  such that

$$\mathbf{A}^\top \mathbf{u} = c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k.$$

Applying  $\frac{1}{\lambda} \mathbf{A}$  to both sides and using part (b),

$$\mathbf{u} = \frac{1}{\lambda} \mathbf{A}(\mathbf{A}^\top \mathbf{u}) = \frac{1}{\lambda} \sum_{i=1}^k c_i \mathbf{A} \mathbf{v}_i,$$

so  $\mathbf{u} \in \text{span}\{\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_k\}$ .

Therefore  $\dim(E_\lambda) = \dim(F_\lambda)$ . Since  $\mathbf{A}^\top \mathbf{A}$  and  $\mathbf{A} \mathbf{A}^\top$  are symmetric, they are diagonalizable, so for each  $\lambda > 0$ , the multiplicity of  $\lambda$  equals the dimension of its eigenspace. Hence the nonzero eigenvalues of  $\mathbf{A}^\top \mathbf{A}$  and  $\mathbf{A} \mathbf{A}^\top$  are the same, including multiplicities.

Finally, if  $\mathbf{A}^\top \mathbf{A} \mathbf{v} = \lambda \mathbf{v}$  with  $\lambda > 0$ , then by part (a),

$$\mathbf{A} \mathbf{A}^\top (\mathbf{A} \mathbf{v}) = \lambda (\mathbf{A} \mathbf{v}),$$

and also  $\mathbf{A} \mathbf{v} \neq \mathbf{0}$  because

$$\mathbf{v} = \frac{1}{\lambda} \mathbf{A}^\top (\mathbf{A} \mathbf{v}).$$

Thus  $\mathbf{A} \mathbf{v}$  is an eigenvector of  $\mathbf{A} \mathbf{A}^\top$  with eigenvalue  $\lambda$ .

## Question 5 (15 points)

Let  $\mathbf{A} \in \mathbb{R}^{m \times n}$  and define the linear mapping  $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}^m$  by

$$\Phi(\mathbf{x}) = \mathbf{A} \mathbf{x}.$$

Let  $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$  be an SVD, with  $\mathbf{U} \in \mathbb{R}^{m \times m}$  and  $\mathbf{V} \in \mathbb{R}^{n \times n}$  orthogonal, and  $\mathbf{\Sigma} \in \mathbb{R}^{m \times n}$  diagonal with singular values  $\sigma_1 \geq \dots \geq \sigma_p \geq 0$  where  $p = \min(m, n)$ . Let  $r$  be the largest index such that  $\sigma_r > 0$  (so  $\sigma_{r+1} = \dots = \sigma_p = 0$  if  $r < p$ ). Denote columns of  $\mathbf{U}$  and  $\mathbf{V}$  by  $\mathbf{u}_i$  and  $\mathbf{v}_i$ , respectively.

Recall that

$$\text{Ker}(\Phi) = \{\mathbf{x} \in \mathbb{R}^n : \Phi(\mathbf{x}) = \mathbf{0}\}, \quad \text{Im}(\Phi) = \{\mathbf{y} \in \mathbb{R}^m : \exists \mathbf{x} \in \mathbb{R}^n \text{ s.t. } \Phi(\mathbf{x}) = \mathbf{y}\}.$$

(a) (7 points) Prove that

$$\text{Ker}(\Phi) = \text{span}\{\mathbf{v}_{r+1}, \dots, \mathbf{v}_n\}.$$

(b) (8 points) Prove that

$$\text{Im}(\Phi) = \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_r\}.$$

**Solution.**

(a) From  $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$ , we have for each  $i \in \{1, \dots, p\}$ ,

$$\mathbf{A} \mathbf{v}_i = \sigma_i \mathbf{u}_i,$$

and for  $i > p$  (if  $n > p$ ),  $\mathbf{v}_i$  corresponds to a zero column of  $\mathbf{\Sigma}$ , so  $\mathbf{A} \mathbf{v}_i = \mathbf{0}$  as well. Hence for every  $i \in \{r+1, \dots, n\}$ ,

$$\Phi(\mathbf{v}_i) = \mathbf{A} \mathbf{v}_i = \mathbf{0},$$

so

$$\text{span}\{\mathbf{v}_{r+1}, \dots, \mathbf{v}_n\} \subseteq \text{Ker}(\Phi).$$

For the reverse inclusion, let  $\mathbf{x} \in \mathbb{R}^n$  satisfy  $\Phi(\mathbf{x}) = \mathbf{Ax} = \mathbf{0}$ . Since  $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$  is an orthonormal basis, write

$$\mathbf{x} = \sum_{i=1}^n c_i \mathbf{v}_i.$$

Then

$$\Phi(\mathbf{x}) = \mathbf{Ax} = \sum_{i=1}^n c_i \mathbf{Av}_i = \sum_{i=1}^r c_i \sigma_i \mathbf{u}_i,$$

because  $\mathbf{Av}_i = \mathbf{0}$  for  $i > r$ . Since  $\{\mathbf{u}_1, \dots, \mathbf{u}_r\}$  is orthonormal and  $\sigma_i > 0$  for  $i \leq r$ , the equality

$$\sum_{i=1}^r c_i \sigma_i \mathbf{u}_i = \mathbf{0}$$

forces  $c_i \sigma_i = 0$  for each  $i \leq r$ , hence  $c_i = 0$  for all  $i \leq r$ . Therefore

$$\mathbf{x} = \sum_{i=r+1}^n c_i \mathbf{v}_i \in \text{span}\{\mathbf{v}_{r+1}, \dots, \mathbf{v}_n\}.$$

This proves

$$\text{Ker}(\Phi) = \text{span}\{\mathbf{v}_{r+1}, \dots, \mathbf{v}_n\}.$$

(b) First show

$$\text{Im}(\Phi) \subseteq \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_r\}.$$

Take any  $\mathbf{y} \in \text{Im}(\Phi)$ . Then there exists  $\mathbf{x} \in \mathbb{R}^n$  such that

$$\mathbf{y} = \Phi(\mathbf{x}) = \mathbf{Ax}.$$

Expand  $\mathbf{x} = \sum_{i=1}^n c_i \mathbf{v}_i$ . Then

$$\mathbf{y} = \mathbf{Ax} = \sum_{i=1}^n c_i \mathbf{Av}_i = \sum_{i=1}^r c_i \sigma_i \mathbf{u}_i \in \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_r\}.$$

For the reverse inclusion, for each  $i \leq r$  we have

$$\mathbf{Av}_i = \sigma_i \mathbf{u}_i,$$

so

$$\mathbf{u}_i = \frac{1}{\sigma_i} \mathbf{Av}_i = \Phi\left(\frac{1}{\sigma_i} \mathbf{v}_i\right) \in \text{Im}(\Phi).$$

Therefore

$$\text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_r\} \subseteq \text{Im}(\Phi).$$

Hence

$$\text{Im}(\Phi) = \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_r\}.$$

## Question 6 (15 points)

Let

$$\mathbf{C} = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

and define the linear map  $\Psi : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  by  $\Psi(\mathbf{x}) = \mathbf{Cx}$ .

- (a) (6 points) Find an orthonormal eigenbasis of  $\mathbf{C}$  and write an orthogonal diagonalization

$$\mathbf{C} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\top.$$

- (b) (5 points) Without computing  $\mathbf{C}^\top \mathbf{C}$ , write down an SVD of  $\mathbf{C}$ .

- (c) (4 points) Use your orthogonal diagonalization to compute

$$\mathbf{C}^k \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \text{for every integer } k \geq 1.$$

**Solution.**

- (a) We first find the eigenvalues and eigenvectors of  $\mathbf{C}$ .

Since

$$\mathbf{C} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 0 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix},$$

one eigenvalue is 4, with eigenvector  $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ .

Also,

$$\mathbf{C} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

so 1 is an eigenvalue, with eigenvector  $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ .

Finally,

$$\mathbf{C} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

so 0 is an eigenvalue, with eigenvector  $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ .

Normalize these eigenvectors:

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{q}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{q}_3 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}.$$

These vectors form an orthonormal eigenbasis. Hence

$$\mathbf{Q} = [\mathbf{q}_1 \quad \mathbf{q}_2 \quad \mathbf{q}_3] = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

and

$$\mathbf{C} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\top.$$

- (b) Since  $\mathbf{C}$  is symmetric positive semidefinite, an SVD can be obtained directly from its orthogonal diagonalization:

$$\mathbf{C} = \mathbf{Q}\mathbf{\Sigma}\mathbf{Q}^\top,$$

where the singular values are the eigenvalues 4, 1, 0. Thus one valid SVD is

$$\mathbf{U} = \mathbf{Q}, \quad \mathbf{\Sigma} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{V} = \mathbf{Q}.$$

- (c) Let

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Using the orthonormal eigenbasis from part (a),

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{q}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{q}_3 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix},$$

we decompose  $\mathbf{x}$  as

$$\mathbf{x} = \frac{1}{\sqrt{2}}\mathbf{q}_1 + \mathbf{q}_2 + \frac{1}{\sqrt{2}}\mathbf{q}_3.$$

Since the corresponding eigenvalues are 4, 1, 0, for every  $k \geq 1$ ,

$$\mathbf{C}^k \mathbf{x} = 4^k \cdot \frac{1}{\sqrt{2}}\mathbf{q}_1 + 1^k \cdot \mathbf{q}_2 + 0^k \cdot \frac{1}{\sqrt{2}}\mathbf{q}_3 = 4^k \cdot \frac{1}{\sqrt{2}}\mathbf{q}_1 + \mathbf{q}_2.$$

Substituting the vectors gives

$$\mathbf{C}^k \mathbf{x} = 4^k \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2^{2k-1} \\ 2^{2k-1} \\ 1 \end{bmatrix}.$$

Equivalently,

$$\mathbf{C}^k \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{4^k}{2} \\ \frac{4^k}{2} \\ 1 \end{bmatrix}, \quad k \geq 1.$$

## Bonus (5 points)

Let  $\mathbf{A} \in \mathbb{R}^{m \times n}$  have SVD

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top,$$

where  $\mathbf{U} \in \mathbb{R}^{m \times m}$  and  $\mathbf{V} \in \mathbb{R}^{n \times n}$  are orthogonal, and  $\mathbf{\Sigma} \in \mathbb{R}^{m \times n}$  has singular values  $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$  on its diagonal. Let  $\mathbf{v}_1, \dots, \mathbf{v}_n$  be the columns of  $\mathbf{V}$ .

Take any unit vector  $\mathbf{x} \in \mathbb{R}^n$ , and write

$$\mathbf{x} = \sum_{i=1}^n c_i \mathbf{v}_i.$$

Prove that

$$\|\mathbf{A}\mathbf{x}\|_2^2 = \sum_{i=1}^n \sigma_i^2 c_i^2,$$

where for indices beyond the diagonal length of  $\Sigma$ , we interpret  $\sigma_i = 0$ .

**Solution.** Since  $\mathbf{V}$  is orthogonal,  $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$  is an orthonormal basis, so

$$\mathbf{x} = \sum_{i=1}^n c_i \mathbf{v}_i \quad \text{and} \quad \sum_{i=1}^n c_i^2 = \|\mathbf{x}\|_2^2 = 1.$$

Also,

$$\mathbf{V}^\top \mathbf{x} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}.$$

Therefore

$$\mathbf{Ax} = \mathbf{U}\Sigma\mathbf{V}^\top \mathbf{x} = \mathbf{U}\Sigma \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}.$$

Therefore

$$\mathbf{Ax} = \mathbf{U}\Sigma\mathbf{V}^\top \mathbf{x} = \mathbf{U}\Sigma \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}.$$

Now use that  $\mathbf{U}$  is orthogonal, so  $\mathbf{U}^\top \mathbf{U} = \mathbf{I}$ . Hence

$$\begin{aligned} \|\mathbf{Ax}\|_2^2 &= \left\| \mathbf{U}\Sigma \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \right\|_2^2 \\ &= \left( \mathbf{U}\Sigma \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \right)^\top \left( \mathbf{U}\Sigma \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \right) \\ &= [c_1 \quad \cdots \quad c_n] \Sigma^\top \mathbf{U}^\top \mathbf{U} \Sigma \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \\ &= [c_1 \quad \cdots \quad c_n] \Sigma^\top \Sigma \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}. \end{aligned}$$

Since  $\Sigma$  is diagonal with diagonal entries  $\sigma_1, \dots, \sigma_n$  (interpreting  $\sigma_i = 0$  beyond the diagonal length), we obtain

$$\|\mathbf{Ax}\|_2^2 = \sum_{i=1}^n \sigma_i^2 c_i^2.$$